ALA 5 | SKILL BUILDING TASK

GMIU

STUDENT RESULT MANAGEMENT SYSTEM

A PHP + MySQL web application for automated marks processing, grading, analytics and printable result cards

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SEMESTER	Sem 5
SECTION	Class A
MAX MARKS	10

GitHub: https://github.com/devachhada2-sudo/student_result_management_system/tree/main/student-result-management

Technology: PHP 8 | MySQL | Bootstrap 5 | Chart.js | XAMPP

A Abstract

Executive summary

The **Student Result Management System (SRMS)** is a web application that replaces error-prone manual grading with an automated, secure digital process. An administrator can register students, enter marks for five subjects and instantly get the total, percentage, grade and PASS/FAIL status. The system adds a colourful dashboard with live charts, a quick search, and an official-style printable result card. It is built only with free technologies: PHP, MySQL, Bootstrap 5 and Chart.js.

Instant Calculation

Total, percentage and grade (A+ to F) are computed automatically.

Live Analytics

Chart.js graphs show grade spread and pass vs fail.

Secure by Design

CSRF tokens, prepared statements, hashed passwords, XSS escaping.

Key Highlights

- 5 subjects, 500 maximum marks, grading from A+ to F
- Admin login with session protection
- Add, edit, view and delete students with confirmation
- Search by enrollment number or name
- Printable result card and class summary report

1. Introduction

Educational institutions manage exams for hundreds of students. Spreadsheets and paper ledgers cause arithmetic mistakes, duplicate records and slow publication. SRMS provides one central portal where results are calculated correctly and retrieved in seconds.

Limitations of the Manual System

- **High error rate:** manual totals often differ
- **Data inconsistency:** records scattered across files
- **No security:** no access control or audit
- **Delays:** students wait long for verified results

2 Requirement Analysis

Hardware, software and feasibility

2.1 Hardware Requirements

Component	Minimum	Recommended
Processor	Dual core 2.0 GHz	Quad core / Intel i5 or better
RAM	2 GB	8 GB or more
Storage	500 MB free	SSD with 2 GB+
Display	1024 x 768	Full HD 1920 x 1080

2.2 Software Requirements

Component	Technology	Purpose
Operating System	macOS / Linux / Windows	Development and server
Backend	PHP 8.x	Server-side logic
Database	MySQL 8 / MariaDB	Data and foreign keys
Web Server	Apache (XAMPP)	Serves the application
Frontend	Bootstrap 5.3 + CSS3	Responsive interface
Charts	Chart.js 4 (CDN)	Dashboard analytics

2.3 Feasibility

Technical

Standard PHP and MySQL, no heavy dependencies.

Economic

100% free and open-source stack, zero licence cost.

Operational

Simple interface that needs almost no training.

3 System Architecture

Three-tier design

SRMS follows a classic **3-tier architecture** that separates presentation, business logic and data storage.

Presentation Tier	Logic Tier	Data Tier
HTML5, Bootstrap 5, CSS3, Chart.js	PHP 8: login, CSRF, grade engine	MySQL: students, results, admins

Design Choices

- **PHP 8:** fast runtime with built-in password hashing
- **MySQL:** foreign keys with ON DELETE CASCADE keep data consistent
- **Bootstrap 5 CDN:** responsive grid without heavy bundles
- **Chart.js:** client-side charts, light on the server

Responsive design: viewport meta tag and flexible layouts give a clean view on desktops, tablets and phones.

4 Database Design

Schema and data dictionary

Database name: **student_result_management**. Three tables: admins, students and results. Each student has exactly one result (1-to-1).

4.1 Table: students

Field	Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Student ID
enrollment_no	VARCHAR(20)	UNIQUE, NOT NULL	Enrollment number
student_name	VARCHAR(100)	NOT NULL	Full name
email / phone	VARCHAR	email NOT NULL	Contact details
course, semester, section	VARCHAR	-	Academic info
created_at	TIMESTAMP	DEFAULT NOW	Created time

4.2 Table: results

Field	Type	Description
student_id	INT, UNIQUE, FOREIGN KEY	References students(id), ON DELETE CASCADE
web_development, database_management, software_engineering, cyber_security, computer_networks	INT (0-100)	Marks of the five subjects
total_marks	INT	Total out of 500
percentage	DECIMAL(5,2)	Percentage
grade / result_status	VARCHAR	A+ to F / PASS or FAIL

Grading Scale	90-100 A+	80-89 A	70-79 B+	60-69 B	50-59 C	40-49 D	Below 40 F

5 System Design and Modules

Workflow and core features

5.1 Data Flow

1. **Login:** admin enters credentials -> password hash verified -> session starts.
2. **Add Student:** details entered -> unique enrollment checked -> saved.
3. **Marks Entry:** marks typed -> JavaScript shows live preview -> PHP calculates and saves.
4. **Result:** search by enrollment or name -> joined data -> printable result card.

5.2 Core Modules

Module	What it does
Authentication	Session login, session ID regenerated, CSRF tokens
Student Registry	Add, edit, view, delete with search and confirmation modal
Live Marks Engine	Total, percentage, grade and PASS/FAIL update while typing
Analytics	Grade distribution bar chart, pass vs fail doughnut, top students
Search and Result Card	Find by name or enrollment, print official result card

6 Security

Protection measures

6.1 CSRF Protection

Every form carries a random 32-character token stored in the session and checked with a timing-safe comparison.

```
function csrf(){ if(empty($_SESSION['t'])) $_SESSION['t']=bin2hex(random_bytes(16)); return $_SESSION['t']; }
function chk(){ if(!hash_equals($_SESSION['t'] ?? '', $_POST['t'] ?? '')) die('Invalid CSRF token'); }
```

- **SQL injection:** all queries use prepared statements
- **XSS:** output escaped with htmlspecialchars(ENT_QUOTES, UTF-8)
- **Passwords:** password_hash() with BCRYPT and password_verify()
- **Sessions:** session_regenerate_id() on login, protected pages check login

7 User Interface

Screenshots of the running application (1/3)

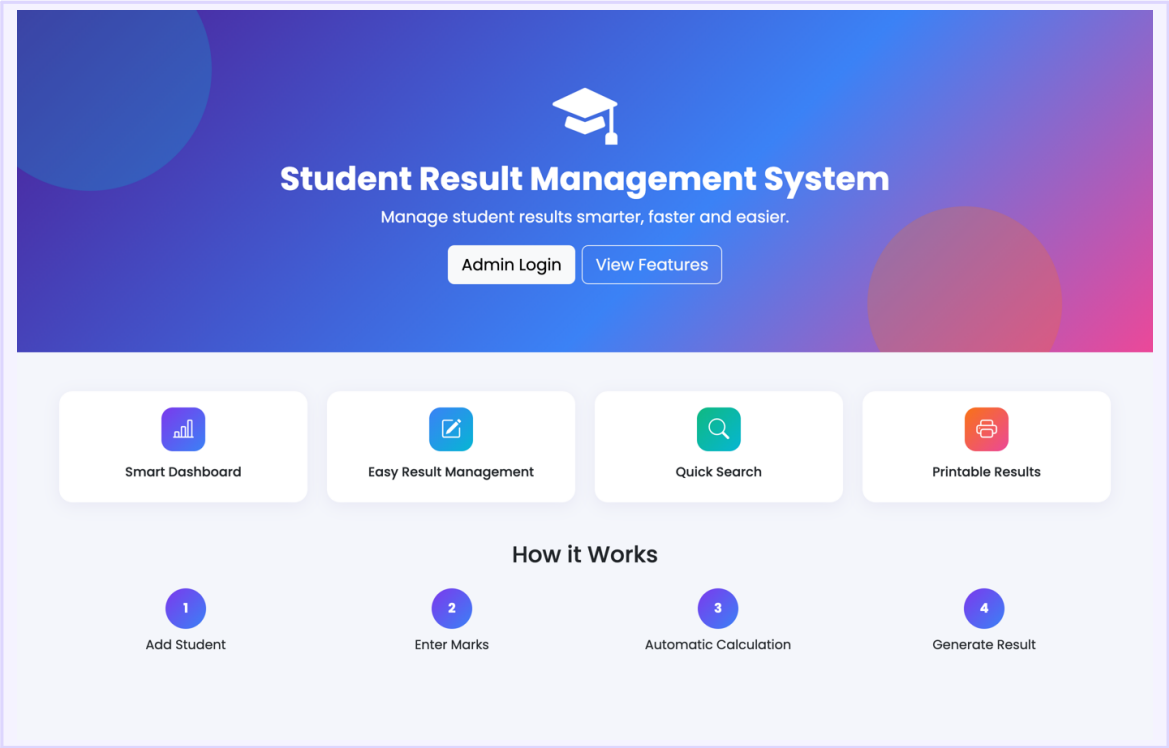


Figure 8.1: Landing page with hero section, feature cards and How it Works

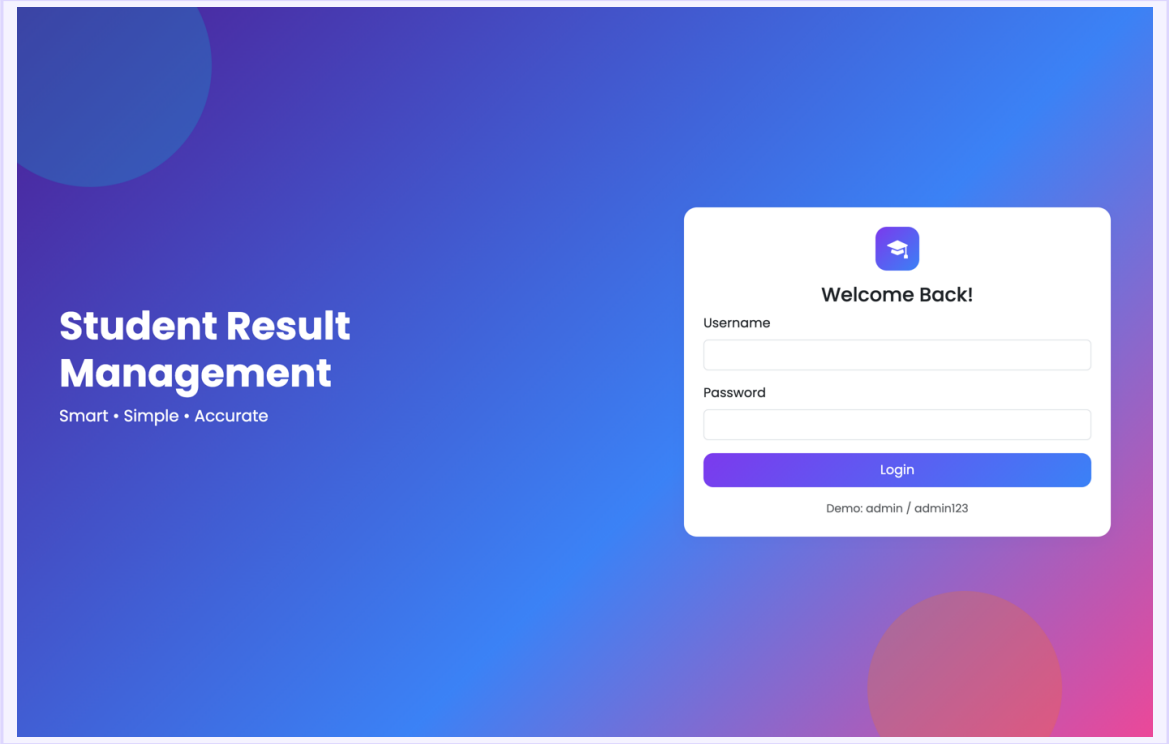


Figure 8.2: Admin login page with demo credentials

7 User Interface

Screenshots of the running application (2/3)



Figure 8.3: Dashboard with statistics, grade chart, pass vs fail chart and top students

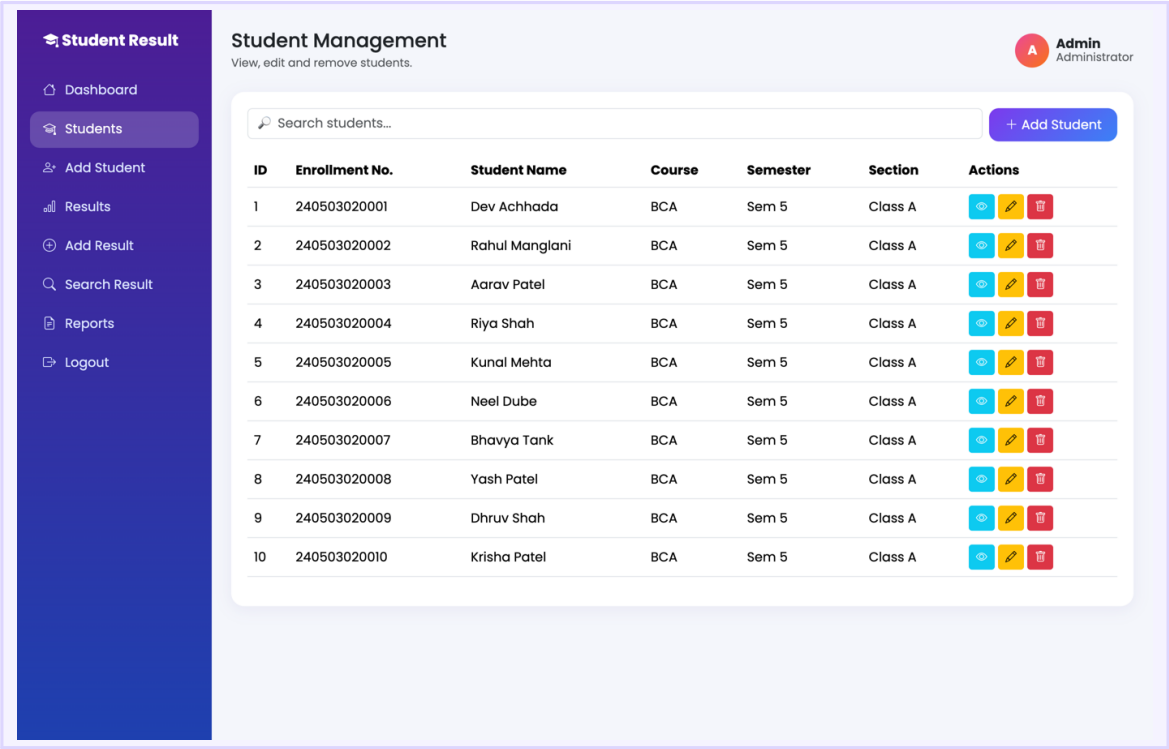


Figure 8.4: Student management table with search, edit and delete

7 User Interface

Screenshots of the running application (3/3)

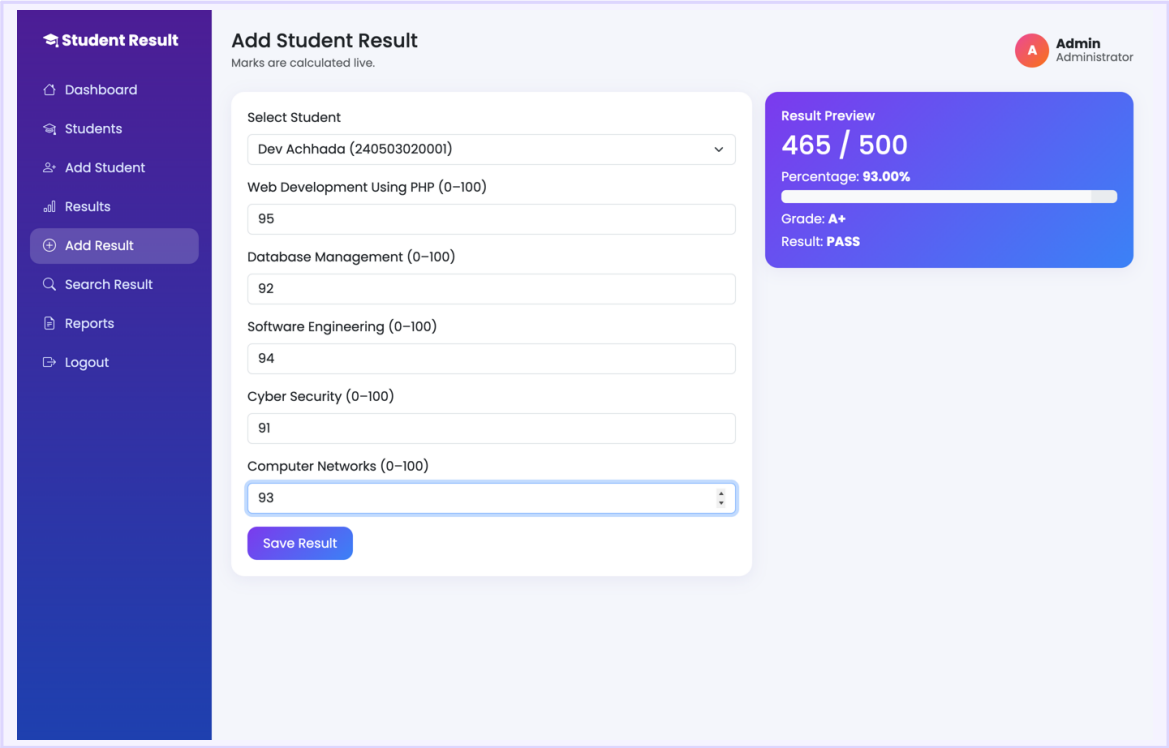


Figure 8.5: Add result page with live total, percentage, grade and result

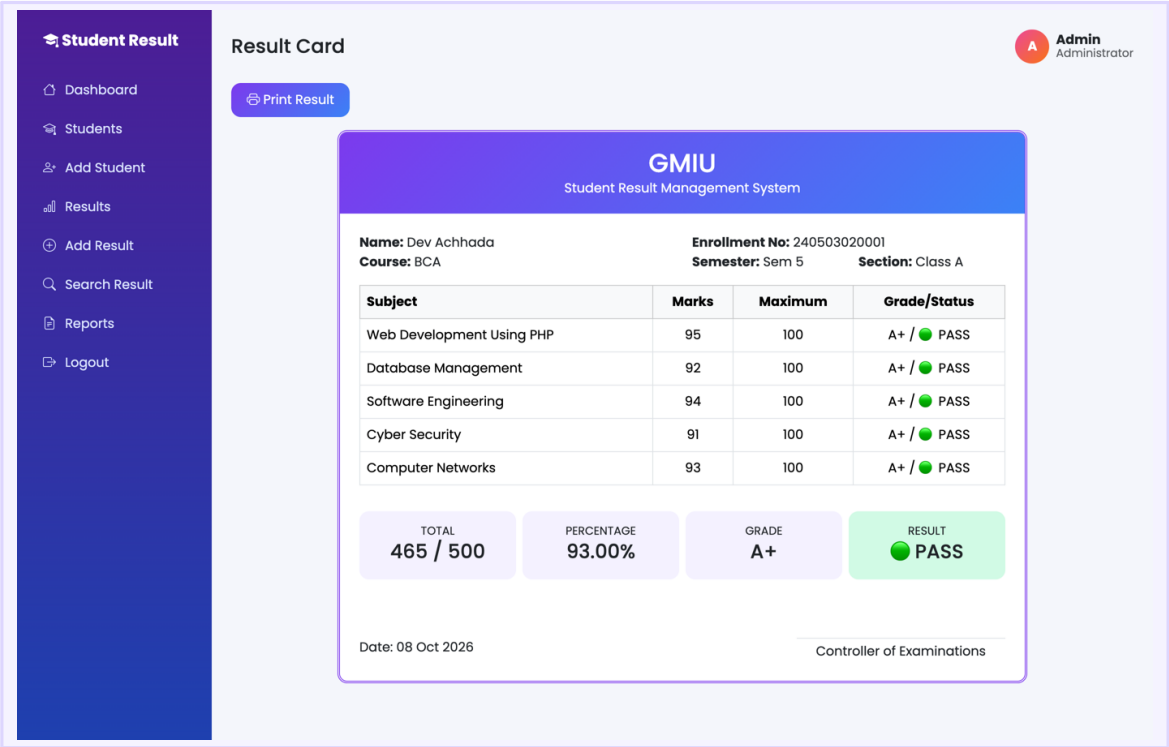


Figure 8.6: Official result card: PASS result with print option

8 Source Code

Core logic

8.1 Grade Calculation (PHP)

```
function grade($p){
  return $p>=90?'A+':($p>=80?'A':($p>=70?'B+':($p>=60?'B':($p>=50?'C':($p>=40?'D':'F'))))));
}
```

8.2 Live Calculator (JavaScript)

```
const mk=document.querySelectorAll('.mk');
function calc(){
  let t=0; mk.forEach(i=>t+=Math.min(100,Math.max(0,+i.value||0)));
  const p=t/5;
  pTot.textContent=t+' / 500';  pPer.textContent=p.toFixed(2)+'%';
  pBar.style.width=p+'%';       pGr.textContent=gradeOf(p);
  pRes.textContent=p>=40?'PASS':'FAIL';
}
mk.forEach(i=>i.addEventListener('input',calc));
```

9 Testing

Test cases

ID	Feature	Input	Expected result	Status
TC-01	Admin login	admin / admin123	Opens dashboard	PASS
TC-02	Invalid login	Wrong password	Error message shown	PASS
TC-03	Add student	New enrollment number	Record saved	PASS
TC-04	Duplicate enrollment	Existing number	Error, nothing saved	PASS
TC-05	Marks validation	105 or negative	Rejected (0-100 only)	PASS
TC-06	Grade calculation	95, 92, 94, 91, 93	465/500, 93%, A+, PASS	PASS
TC-07	Search	Dev Achhada	Result link shown	PASS
TC-08	CSRF check	POST without token	Request blocked	PASS

10 Deployment Guide

Run the project locally

XAMPP / Apache

- Copy the folder to **htdocs/student-result-management**
- Start Apache and MySQL from the XAMPP Control Panel
- Open phpMyAdmin and import **database/student_result_management.sql**
- Open http://localhost/student-result-management/
- Login with **admin / admin123**
- Source code on GitHub: https://github.com/devachhada2-sudo/student_result_management_system/tree/main/student-result-management

11 Conclusion and Future Scope

Summary

SRMS meets all ALA 5 requirements: marks entry, automatic total and percentage, grade display and result search. It removes calculation mistakes, saves time and presents results in a clean, modern interface built entirely with free tools.

Future Enhancements

- Separate login for students, faculty and super admin
- SMS or email notification when results are published
- Server-side PDF marksheet with QR verification
- Semester-wise and multi-year result history

References

No.	Source	Link
1	PHP Manual	php.net
2	MySQL Reference Manual	dev.mysql.com
3	Bootstrap 5.3 Documentation	getbootstrap.com
4	OWASP Top 10	owasp.org
5	Chart.js Documentation	chartjs.org
6	Project Source Code (GitHub)	github.com/devachhada2-sudo